

Task-Oriented Semantic Granularity Selection for Bandwidth-Constrained V2V Cooperative Perception

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Abstract—Cooperative perception must choose what to transmit, not only how much to compress. We formalise per-frame semantic granularity selection over three actions—ego-only (E), object-level (L) and feature-level (F)—and evaluate it under a single detector, a single field of view and a single ground truth, with communication charged through a measured LDPC/QAM chain rather than a declared budget. The resulting method, CA-TOSG (channel-aware task-oriented semantic granularity), reduced realised communication payload by 99.85% on Test and 99.70% on Culver-City relative to the frozen comparator. However, the preregistered scene-level non-inferiority criterion was not met on either split, indicating that substantial communication reduction was achieved without statistically establishing accuracy preservation. Three mechanism results follow from the unified branch: partial recovery is necessary, since all-or-nothing delivery of a 12,567-codeword message is effectively dead at realistic loss rates; where a loss falls changes the task outcome even when how much is lost is held exactly fixed; and fusing a zeroed feature tensor is not equivalent to ego-only inference. In the frozen policy the feature action is never selected, and we show this follows from the cost structure and the sampled penalty grid rather than from a failure to learn.

Index Terms—Cooperative perception, V2V communication, semantic communication, bandwidth adaptation.

I. Introduction

COOPERATIVE perception lets a vehicle see what its own sensors cannot. What it should send is not settled: raw points, intermediate features and object lists differ by orders of magnitude in size [1], [2] and in what they preserve. We treat the choice itself as the object of study.

Our contributions are three.

- 1) Problem formalisation. Per-frame granularity selection as a budget-constrained action choice over {E, L, F}, with payload charged per frame through the transport chain the messages actually traverse.
- 2) A receiver-driven method. The ego evaluates its own scene and channel and requests a granularity; the selector’s inputs are strictly ego-local and available before the request.
- 3) Experimental findings under a unified protocol. One checkpoint, one field of view, one ground truth, and a deterministic evaluation pipeline.

The random forest used below is one instance of the selection layer, not the contribution. What is being offered is the granularity-control formulation and the accounting protocol that makes a payload claim checkable; any policy that consumes ego-local cues and emits a request slots

into the same framework, and we report below that this particular instance does not beat a tuned threshold.

II. Method

A. Action space and the receiver-driven request

The ego selects one of three actions per frame and signals it with a two-bit request. That request is a fixed control overhead, identical for all three actions, and is therefore not charged to any action’s payload. E transmits nothing. L transmits the collaborator’s detected boxes. F transmits the collaborator’s quantised bottleneck tensor.

B. Communication accounting

Payload is measured, not declared. For F the transmitted tensor is 739,200 elements at $w = 8$ bits, packetised at 8,000 payload bits with a 320-bit header following common V2X framing assumptions [3], [4], giving $N_{cw} = 12,567$ codewords and $B_F = 3.14175$ Msym/frame. B_F is derived from the codeword count rather than from $(\text{info} + \text{header})/R/\log_2 M$, because the direct route ignores codeword padding. L is charged per frame through the identical chain from its own box count.

C. Transport

The message is fragmented into N_{cw} codewords; each succeeds or fails at the frame’s SNR, and the receiver reconstructs from what arrives, zero-filling failed blocks. All-or-nothing delivery is retained as a pessimistic sensitivity arm. Three regimes are evaluated: fragment-aware recovery, packet-level loss, and all-or-nothing message delivery (Fig. 3).

D. Selector inputs

The cue vector has 23 dimensions and is ego-local by construction: 19 statistics of the ego’s own sweep, its own detected box count, a binary collaborator-availability flag, and two channel fields. Every field is computable before the request is issued, so no input depends on a message the ego has not yet asked for.

III. Experimental Protocol

All experiments use OPV2V [5] with a PointPillars detector [6]. One checkpoint serves all three actions; the field of view and ground truth are identical across them. Evaluation is deterministic: the per-sample point shuffle is seeded on (split, scene, frame, cav), and two independent

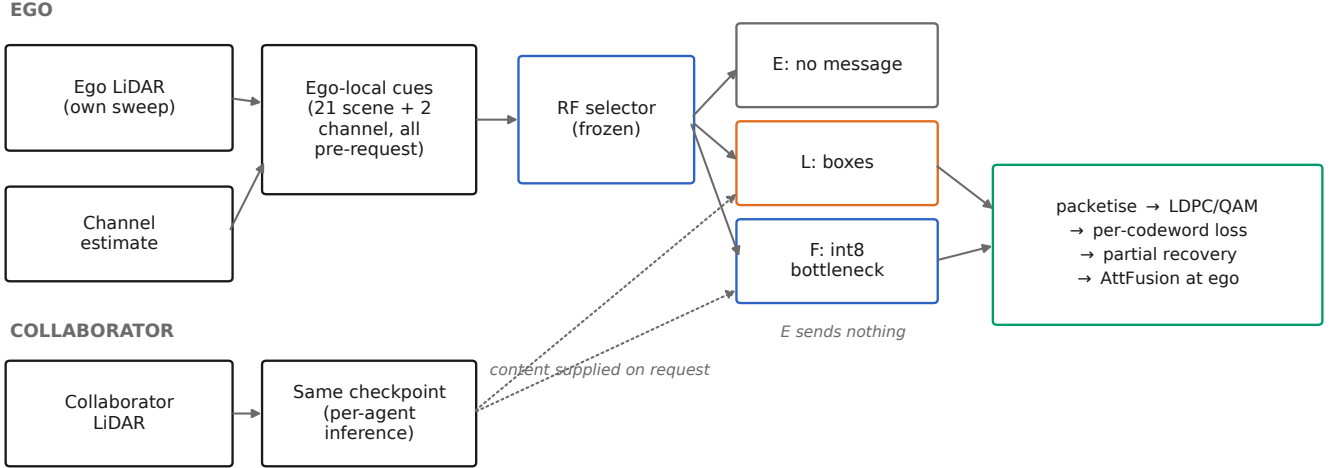


Fig. 1. Data flow as implemented. The cue vector and the channel estimate are ego-side and available before the request; the collaborator supplies message content only when asked. E issues no message and therefore never enters the transport chain.

processes reproduce a run bit for bit. The selector is fitted on the development split only and frozen—a random forest [7], candidate 67 at $\lambda = 0.2$ —before either held-out set is opened. The comparator is likewise frozen: an SNR threshold at $\tau = 16.5$, the best budget-feasible threshold at the primary cell. The confirmatory unit is the scene, not the frame: frames within a scene are consecutive samples of the same traffic and are far from independent. Intervals are scene-level bootstraps over 16 Test scenes and 4 Culver-City scenes, covering 47,740 and 12,100 evaluated rows respectively.

IV. Results

A. Primary comparison

Table I gives the primary cell. On Test the payload falls by 99.85% and on Culver-City by 99.70%; on both splits the scene-level bootstrap lower bound on ΔF_1 lies outside the preregistered margin of -0.005 , so accuracy non-inferiority is not established. Both halves are stated together throughout: the saving is large—a factor of 669 and 339—and the shortfall on the bound is 0.0024 (Test) and 0.0004 (Culver). The Culver-City interval rests on only 4 scenes and is correspondingly coarse; it reproduces the shape of the Test result on much less scene diversity rather than independently corroborating it.

B. Action mix

The frozen selector requests E on 40.3% of Test rows and L on 59.7%, and never requests F ($\rho_F = 0.000$); the Culver-City mix is 40.7% and 59.3%. Section V explains why this is a property of the sampled penalty grid rather than evidence that the feature action is useless.

C. Baselines

Fixed-E, Fixed-L and Fixed-F bound the action space. The internal decision rules are the SNR threshold and

TABLE I
Primary cell. Δ is CA-TOSG minus the frozen comparator; the criterion is defined on the bootstrap bound, not the point estimate.

	Test		Culver-City	
	CA-TOSG	$\tau = 16.5$	CA-TOSG	$\tau = 16.5$
Scene-eq. F_1	0.87679	0.88178	0.87352	0.87813
Payload (Msym)	0.00081	0.54102	0.00147	0.49850
ΔF_1	-0.00499		-0.00461	
ΔF_1 LCB	-0.00738		-0.00541	
Payload saving	99.85%		99.70%	
Scenes	16		4	

two- and three-scalar hand rules; the frozen $\tau = 16.5$ is the primary comparator because it is the only arm sharing CA-TOSG's complete payload definition. Where2comm [1] is included as an external feature-content baseline. It is evaluated under the same data splits, field of view, ground truth and detection metrics. We report its native communication rate because the evaluated implementation transmits floating-point selected features and does not execute the locked int8 quantisation path. Consequently, it is excluded from bit-level budget-matched claims. At a communication threshold of 0.02 it reaches AP@0.5 0.91708 on Test and 0.83195 on Culver-City while retaining 5.2% and 9.1% of features. No adaptive-policy baseline met the reproducibility bar.

V. Limitations

Accuracy non-inferiority was not established. On both splits the point estimate lies inside the margin and the confidence bound does not. The margin was fixed in advance and the bound is what it tests.

Proportion. The saving is overwhelming; the shortfall is small. Neither may be reported without the other: stating only the saving packages a failed criterion as success, and stating only the shortfall discards a 669-fold reduction.

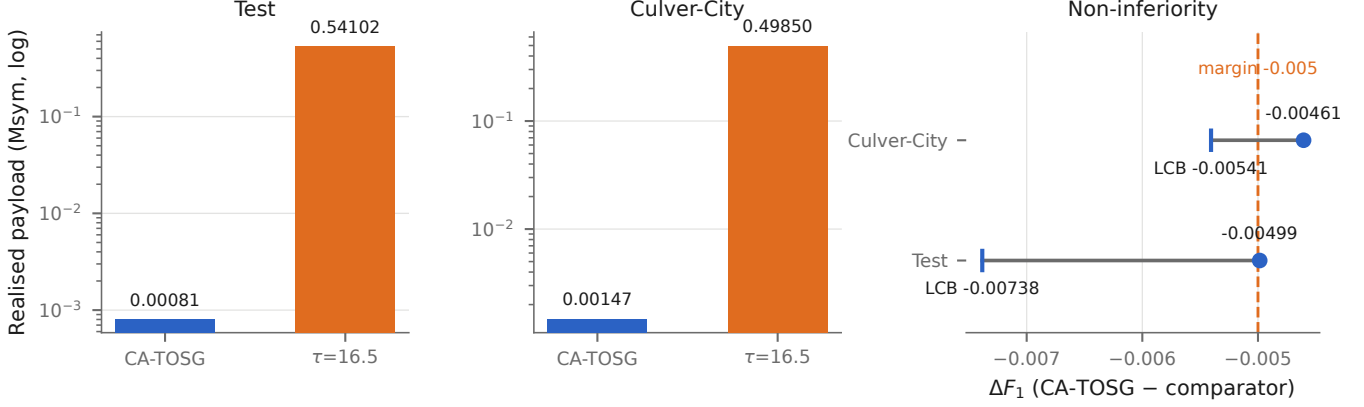


Fig. 2. Primary cell. Payload is drawn on a logarithmic axis and ΔF_1 in its own panel: the payload ratio is 669-fold while the F_1 shortfall on the bound is 0.0024, so a shared scale would erase one half of the result. The right panel puts the preregistered margin and the bootstrap lower bound on one ruler. On both splits the point estimate lies inside the margin and the bound does not.

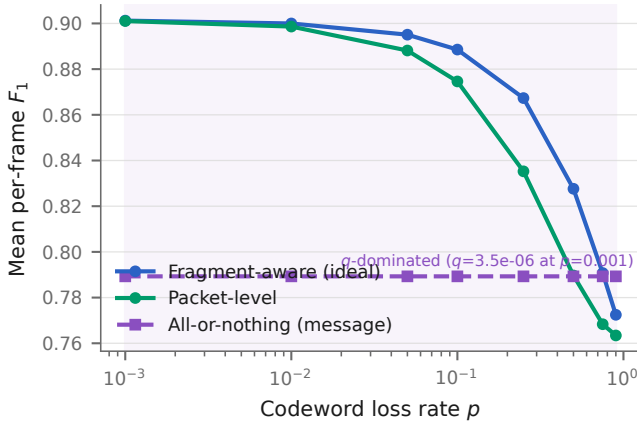


Fig. 3. Delivery regimes on the development split. Fragment-aware partial recovery degrades gracefully; all-or-nothing message delivery is governed by the survival probability q rather than by how the feature action performs.

Full-frame and collaborator-available payload savings differed by 0.000 percentage points on both held-out splits.

Culver-City rests on 4 scenes. The bootstrap resamples them, so its interval is coarse; the result has the same shape as Test on far less scene diversity. Culver-City is a simulated domain shift.

A. Why this remains a three-action method although $\rho_F = 0$

Four measurements belong together. First, F is genuinely the best action on 56.2% of grid rows: its realised utility exceeds both E and L there, so this is not an action that never helps. Second, the break-even λ falls in a gap in the sampled penalty grid—0.01556 conditional on the rows where F is best, and 0.0048 averaged over the whole grid, each value carrying its conditioning set—while the grid samples $\lambda \in \{0, 0.02, 0.05, \dots\}$, so no candidate could have been selected in the region where a mixed policy exists. Third, a finer λ scan, registered before it produced

any number, shows a genuine three-action trade-off across that whole interval, with ρ_F falling smoothly from 0.534 to 0.152. Fourth, all of this is an exploratory diagnostic on the development split: it enters no primary criterion, and the frozen candidate was not replaced on the strength of it. Taken together, $\rho_F = 0$ is a property of where the penalty grid sampled, not evidence that the feature action is worthless.

No three distinct budget points. All three budgets select the same candidate and the budget never binds. The contribution is therefore a granularity-control framework, not a demonstration that a learned selector beats simple rules—at equal budget it does not.

Other limits. Selector-only latency is measured; total system latency is not. One ego–collaborator pair. A flat block-fading channel.

VI. Conclusion

Three transmission granularities were placed under one detector, one field of view and one ground truth, and charged through one measured transport chain. Communication falls by a factor of 669 on Test and 339 on Culver-City relative to a budget-feasible threshold policy, while accuracy non-inferiority is not established on either. The conditions are one dataset family, one detector, a single ego–collaborator pair, and a flat block-fading channel.

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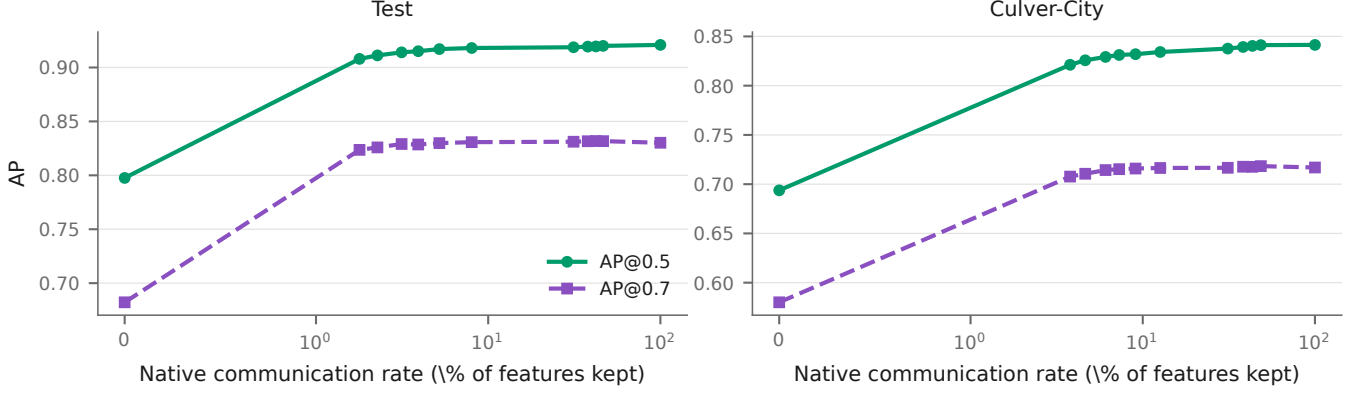


Fig. 4. Where2comm on its native communication-rate axis. The horizontal axis is the fraction of features retained, not a bit budget.

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